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Econometrics Midterm

Preliminary steps:

Stock market and country:
The SMI, the main stock market index of the SIX Swiss Exchange in Switzerland (Zürich).

Name and exact ticker of the index:
- Swiss Market Index (SMI)
- Yahoo Finance ticker: ^SSMI

Initial and final dates of the sample:
- Initial date: January 4, 1999
- Final date: December 30, 2024 (last trading day of 2024 available on Yahoo Finance for this index)

Sample length (number of days/months/years):
- Daily observations: 6,530 trading days recorded
- Monthly observations: 312 months recorded (at the end of each month, from Jan 1999 through Dec 2024)
- Yearly observations: 26 years recorded (from 1999 through 2024)

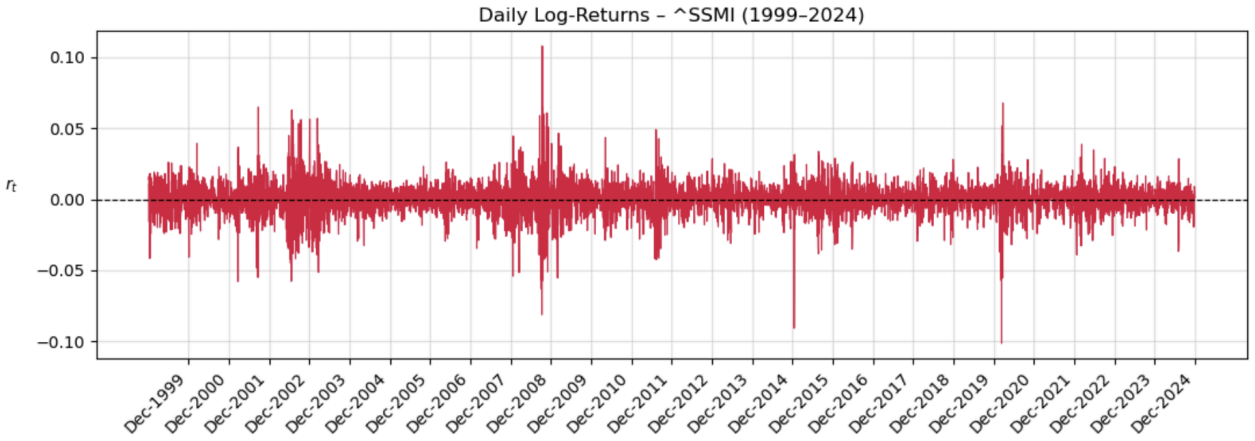
Note: rounded numbers are used throughout the work for clarity, without affecting the substance of the analysis.

	Daily	Weekly	Monthly	Annual
Mean	0.006800	0.030864	0.153323	1.707506
St.Deviation	1.113085	2.465198	3.799528	17.475928
Diameter.C.I.Mean	0.027000	0.131213	0.422285	6.850564
Skewness	-0.268988	-1.211971	-0.632130	-0.832083
Kurtosis	10.581989	16.760345	3.692097	3.246403
Excess.Kurtosis	7.581989	13.760345	0.692097	0.246403
Min	-10.133930	-25.201562	-14.034659	-42.723436
Quant.5%	-1.711129	-3.747698	-6.819685	-30.850651
Quant.25%	-0.527363	-1.116163	-1.858788	-7.025087
Median.50%	0.053740	0.228928	0.674852	3.735616
Quant.75%	0.565208	1.324582	2.651304	14.709786
Quant.95%	1.605006	3.289371	5.324162	22.154314
Max	10.787597	16.288479	10.605297	28.675505
Jarque.Bera.stat	15717.471193	11030.075781	26.919036	2.948088
Jarque.Bera.pvalue	0.000000	0.000000	0.000001	0.228998
Lillie.test.stat	0.069199	0.083356	0.093384	0.105021
Lillie.test.pvalue	0.001000	0.001000	0.001000	0.666926
N.obs	6529.000000	1356.000000	311.000000	25.000000

Question 1:

Fact #2 (Returns Are Stationary):
We want to demonstrate: Returns fluctuate around a stable mean and the variance is roughly constant.

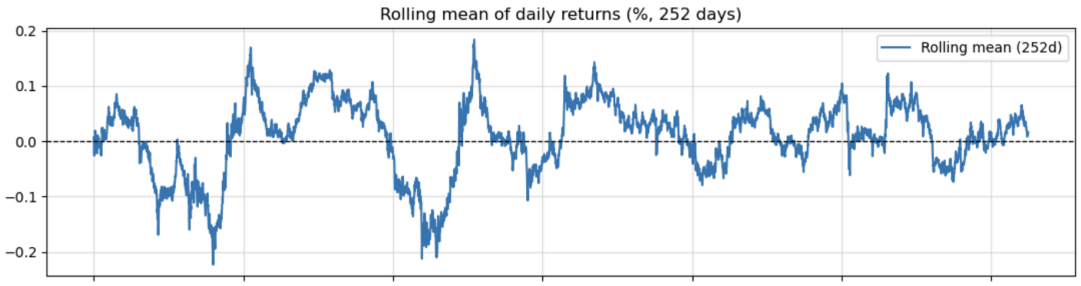
First, with an intuitive approach:



Intuitively, looking at this graph of daily log-returns, the daily returns appear to hover around 0, and are typically contained between [-0.02, 0.02] except for certain exceptions (around 2002, 2008, 2014 and 2020). Which fits the definition of the stationarity of the returns.

Diving into the numbers:
Weak stationarity can be proven if and only if all of the three following criteria are met simultaneously:

- **E(returns) = constant**:



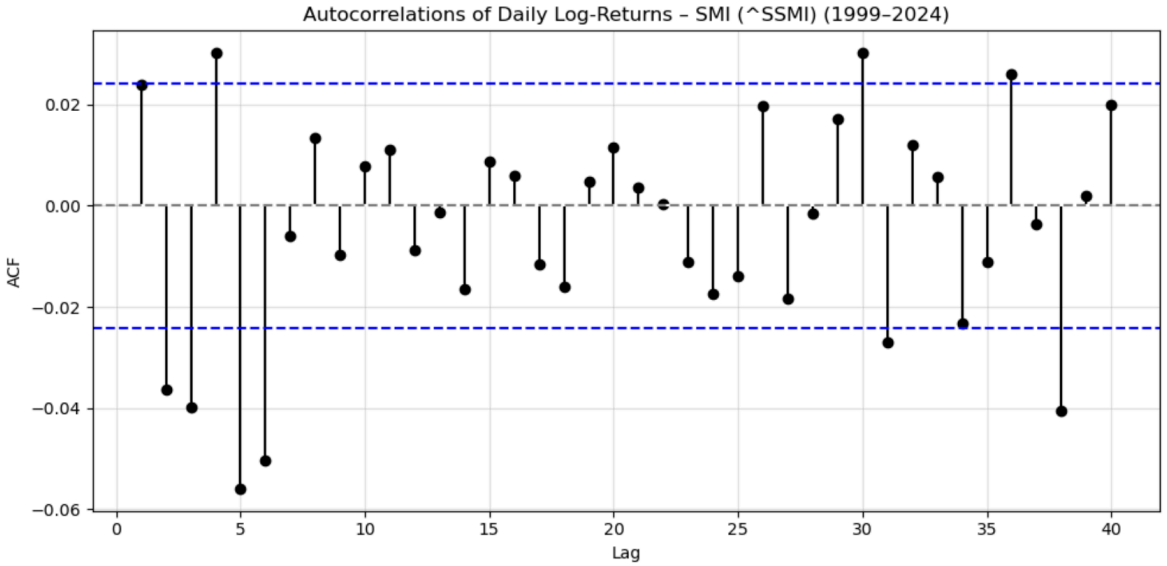
The rolling mean of daily returns stays close to zero without drifting sustainably. That would be the constant mean.

- **Var(returns) = constant:**



The rolling standard deviation shows spikes around 2002-2003, 2009, and 2020. Outside of these crisis periods, it stays within [0.5, 1.25], suggesting stable volatility in normal times because it's a fairly narrow interval. For simplicity, we can take σ = 1.25 as a conservative upper bound to approximate the variance as fairly constant.

- **Cov(r_t,r_t-k) = ɣ_k depends only on the lag k.**



This graph shows that for most lags, the autocorrelation is contained within [-0.02, 0.02]. According to the criterion given in the lecture (under 0.02 is normal, between 0.2 and 0.5 is big, above 0.9 is massive), these autocorrelations are small. The autocorrelation function bars oscillate around zero and remain low across lags, which is consistent with weak stationarity, where the autocovariance only depends on the lag.

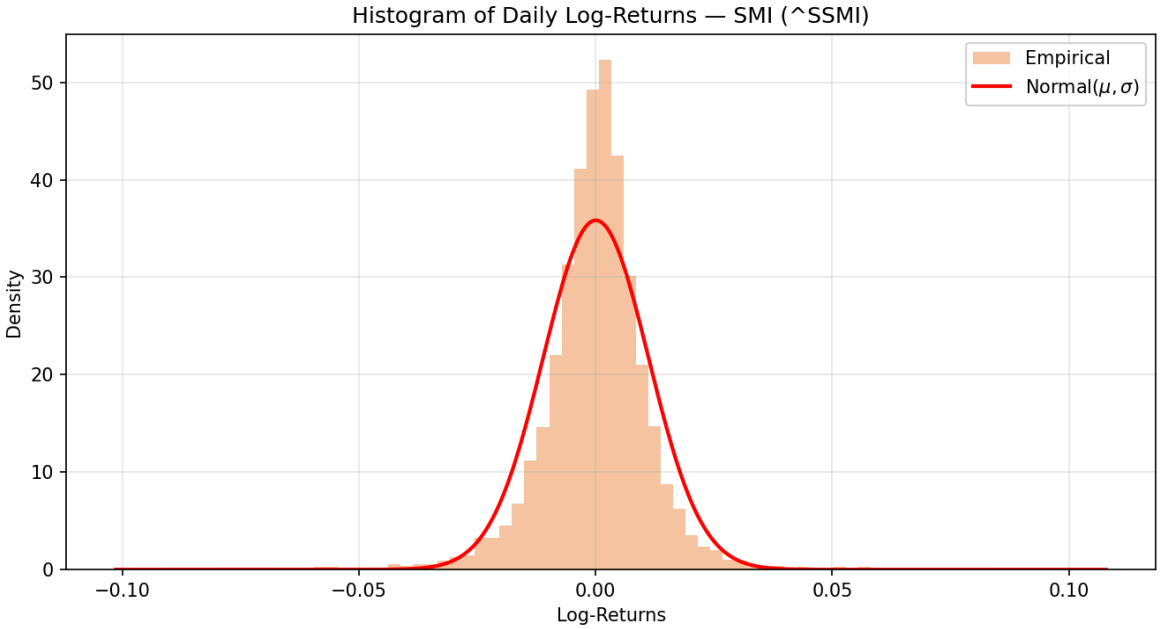
Conclusion: All three criteria are simultaneously met => the SMI data are consistent with a weak stationarity from 1999 through 2024.

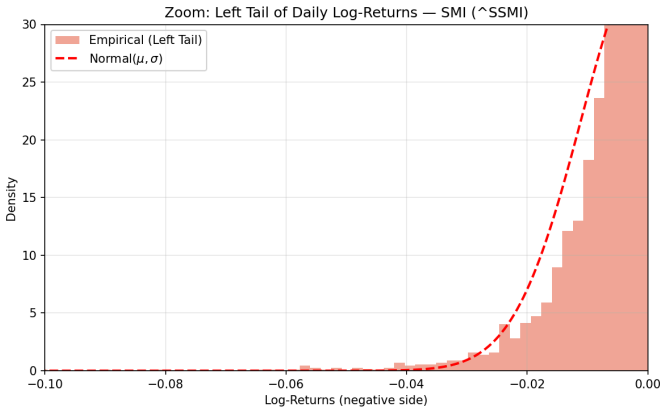
Fact #3 (Negative Skewness):
We want to demonstrate: Price crashes happen more often and on a larger scale than big price surges.

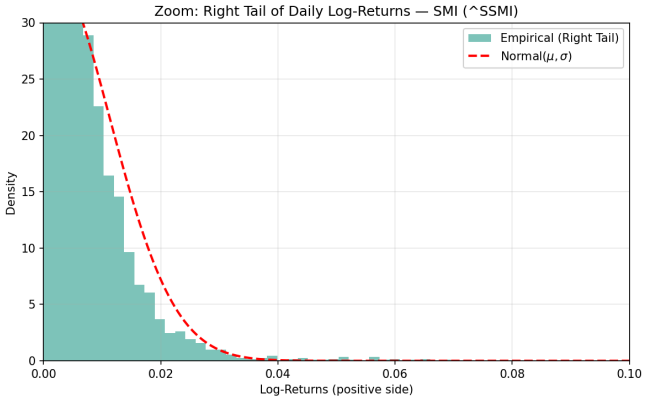
Python gives us:

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Skewness (SMI daily log-returns) = -0.2690
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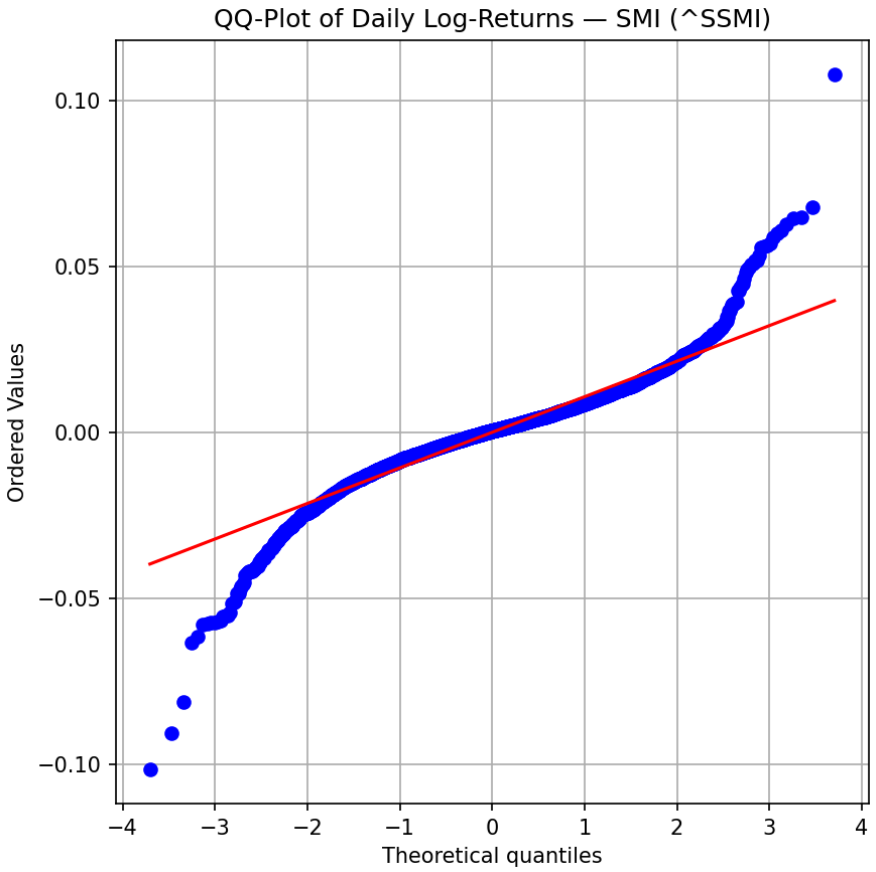
-0.27 < 0 => skewness is negative, therefore returns are negatively skewed.
To confirm it graphically, we know a negative skewness can be witnessed if for r > 0, density(-r) ≥ density(r).







The two smaller graphs are a zoomed-in version of the previous global one. Here, take for example:
density(-0.01) ≈ 17 and density(0.01) ≈ 16
density(-0.02) ≈ 4 and density(0.02) ≈ 3
=> the histograms concur with the computed skewness value being negative.



On the QQ-plot, points (blue, from the SMI) in the lower quantiles lie well below the straight line (the normal distribution), while upper quantiles points lie above it by a noticeably smaller margin. In other words, the SMI deviates more from the normal distribution downward than upward. This is negative skewness.

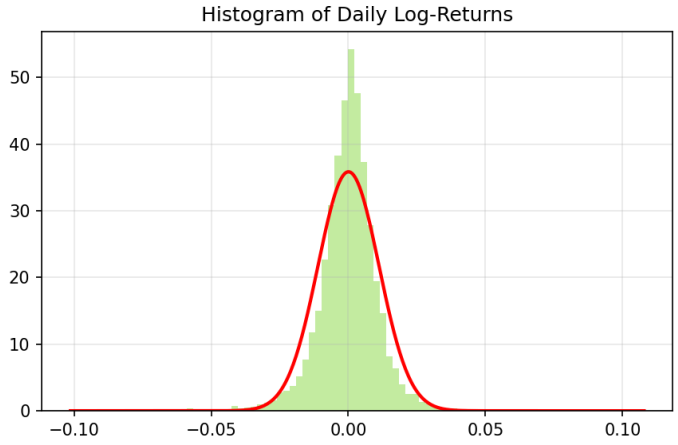
Fact #4 (Heavy Tails):

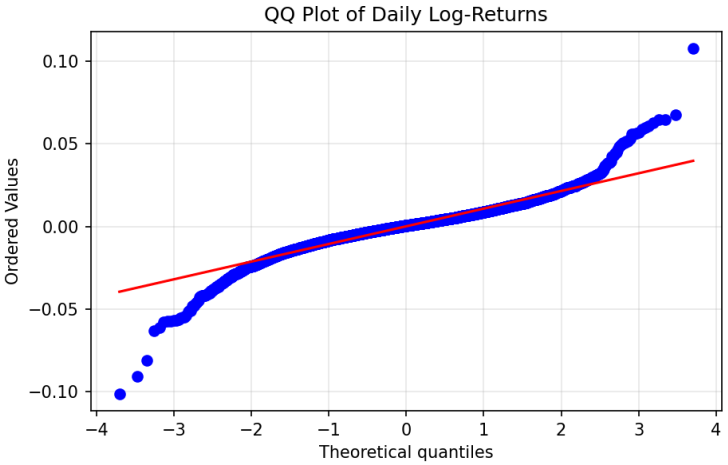
We want to demonstrate: Extreme returns (crashes or rallies) occur more often than in a normal distribution.

Heavy tails are captured by excess kurtosis. The larger the excess kurtosis, the heavier the distribution compared to the normal distribution (instead of it being flat, it'll be spiky and concentrated, typically with a sharper center and fatter tails).

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---- Excess Kurtosis
Daily   : 7.589
Monthly : 0.723
Annual  : 0.588
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The computed excess kurtosis for daily returns is 7.589 > 0.





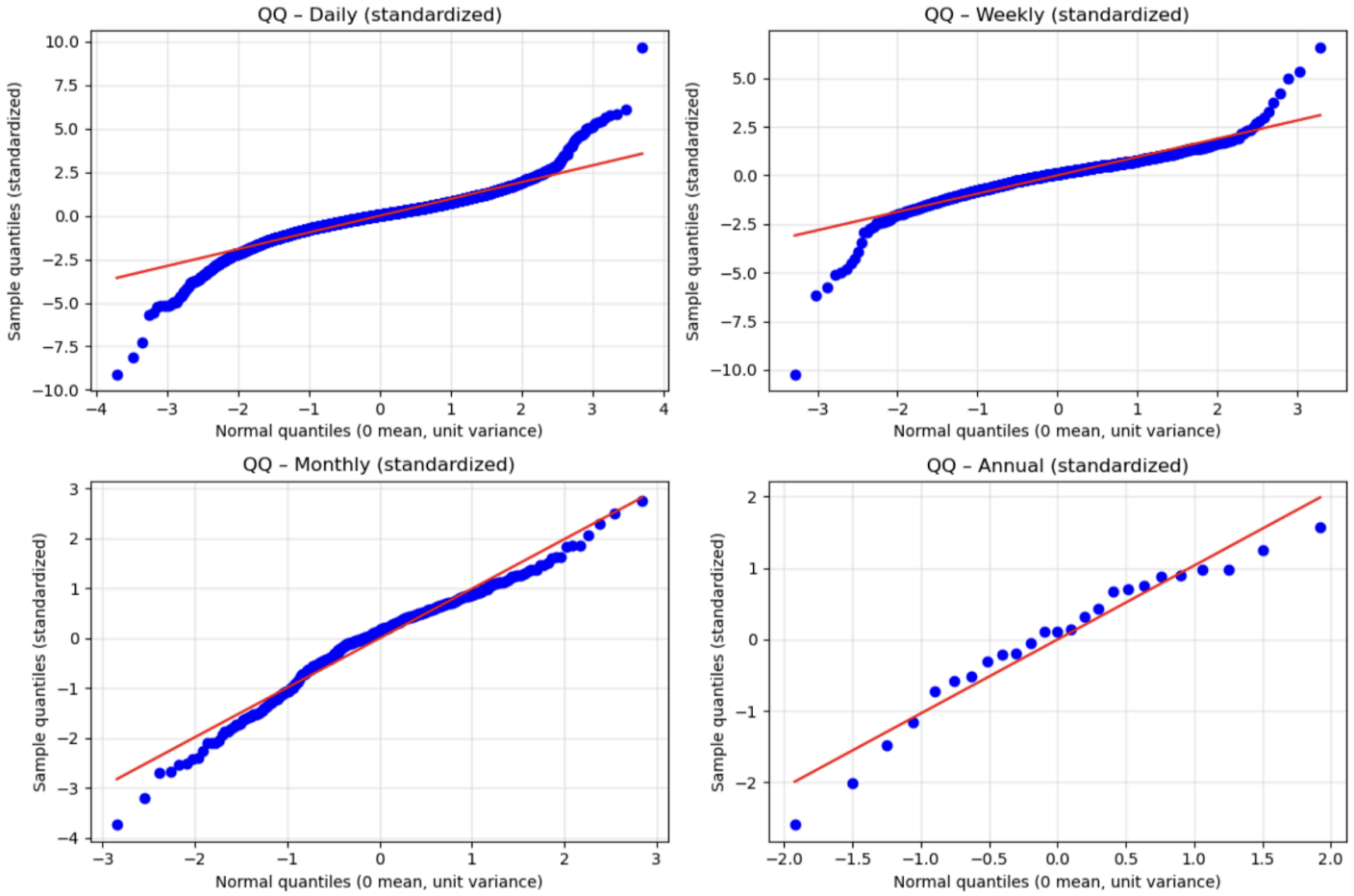
This is confirmed graphically: The histogram shows a sharper center than the normal distribution (in red). And in the QQ-plot, the SMI greatly diverges from the normal distribution in the lower and upper quantiles. Because there's more data than the normal in the extremes, the tails are spikier (fatter) than the normal distribution, which is typical of positive excess kurtosis.

Conclusion: The SMI daily returns show heavier tails than a normal distribution.

Fact #5 (Aggregational Gaussianity):

We want to demonstrate: Over longer periods (monthly, yearly), returns look more normally-distributed than daily returns.

By taking a look at the QQ-plots that follow, one can see that:



At daily and weekly frequencies, the witnessed distribution of the SMI is different from that of a normal distribution because it deviates greatly in the lower and upper quantiles. However, at the monthly and yearly frequencies, the witnessed distribution of the SMI gets much closer to the normal distribution. That's because the longer the period, the more the random (idiosyncratic) variations even out each other, leaving only the underlying trend, which is expected to be a normal distribution.

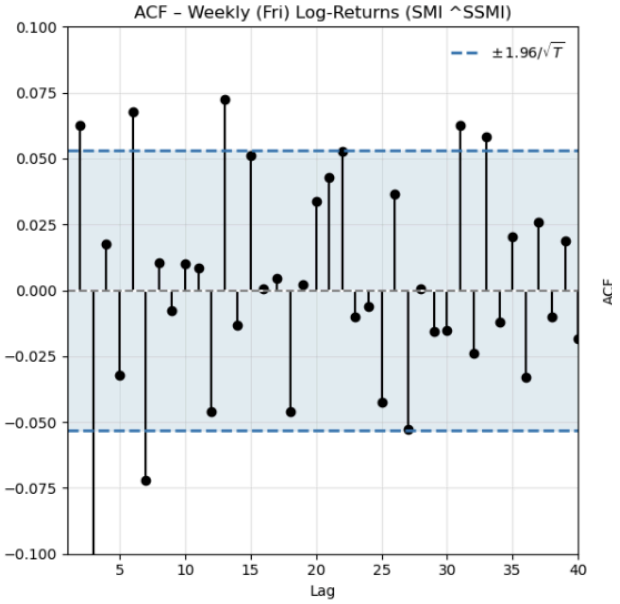
Freq	Skewness	ExcessKurtosis
Daily	-0.269	7.582
Weekly	-1.212	13.760
Monthly	-0.632	0.692
Annual	-0.832	0.246

Furthermore, the computations show that the longer the period, the more the excess kurtosis approaches zero. In other words, as the frequency increases, the SMI kurtosis converges to the normal kurtosis. Which is yet another indicator of the normality of the SMI returns over long periods (because if it's a normal distribution, then it has a normal kurtosis as well).

Fact #6 (No Autocorrelation):

We want to demonstrate: Past returns don't predict future returns.

If there is no autocorrelation, then the autocorrelation function (denoted by "ACF") should fall within the insignificance interval, which at 95% is [-1.96/sqrt(T), 1.96/sqrt(T)], where T is the number of time observations. This interval is represented in light blue on the graphs.

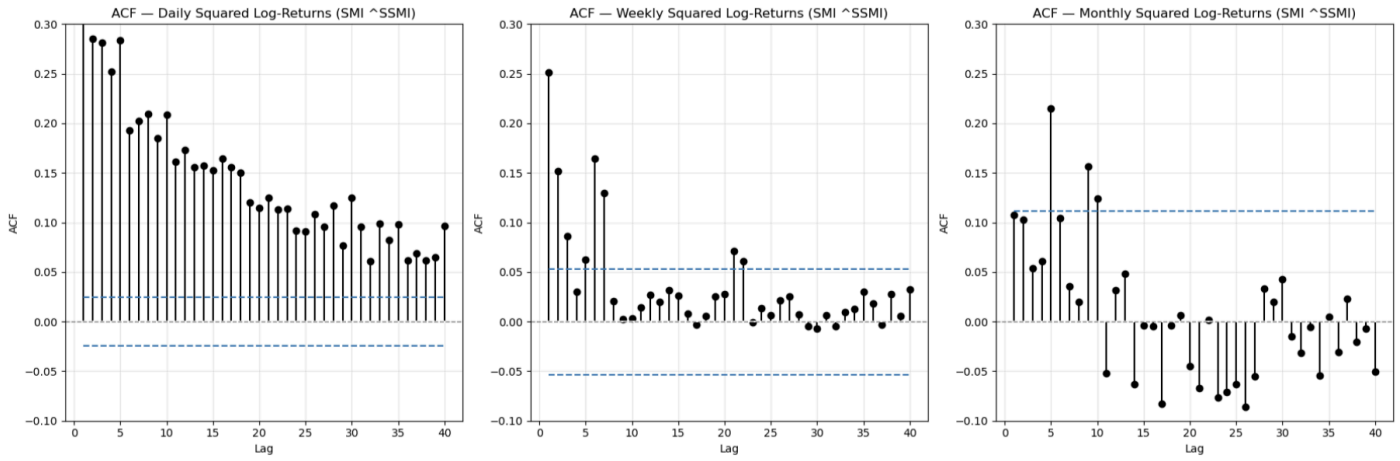


Here, for most lags, the autocorrelation functions do indeed fall inside the light blue zone: therefore, at 95%, there is no autocorrelation between returns for most lags.
This means that past returns aren't correlated with future returns, therefore don't help predict returns.

Fact #7 (Volatility Clustering):

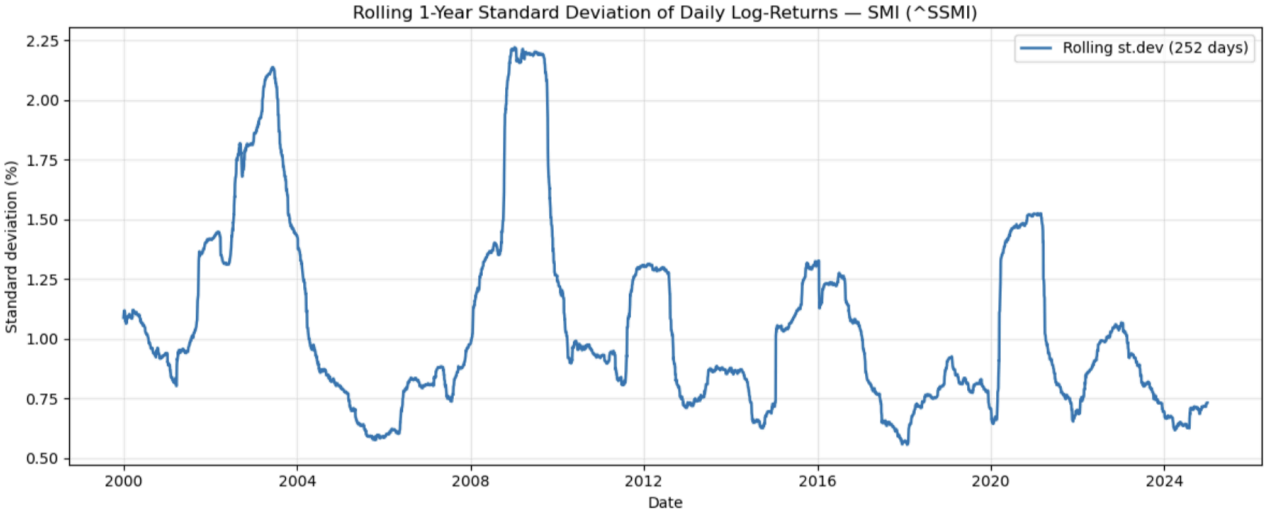
We want to demonstrate: Markets alternate between quiet and stormy periods. When volatility is high one day, it's likely to stay high for a while; when it's low, it stays low.

The following graphs show the autocorrelation functions in sheer magnitude for returns (shown by squared returns). Like for fact #6, there is no autocorrelation if the values fall between the blue broken lines.



At the daily frequency, there clearly is autocorrelation because all values are outside of the boundaries of the blue interval (all are statistically significant. In real life, it means when today is troubled, tomorrow is also likely to be troubled). And autocorrelation remains positive, though decaying slowly across lags, indicating volatility lingers and each day isn't independent (the clustering).

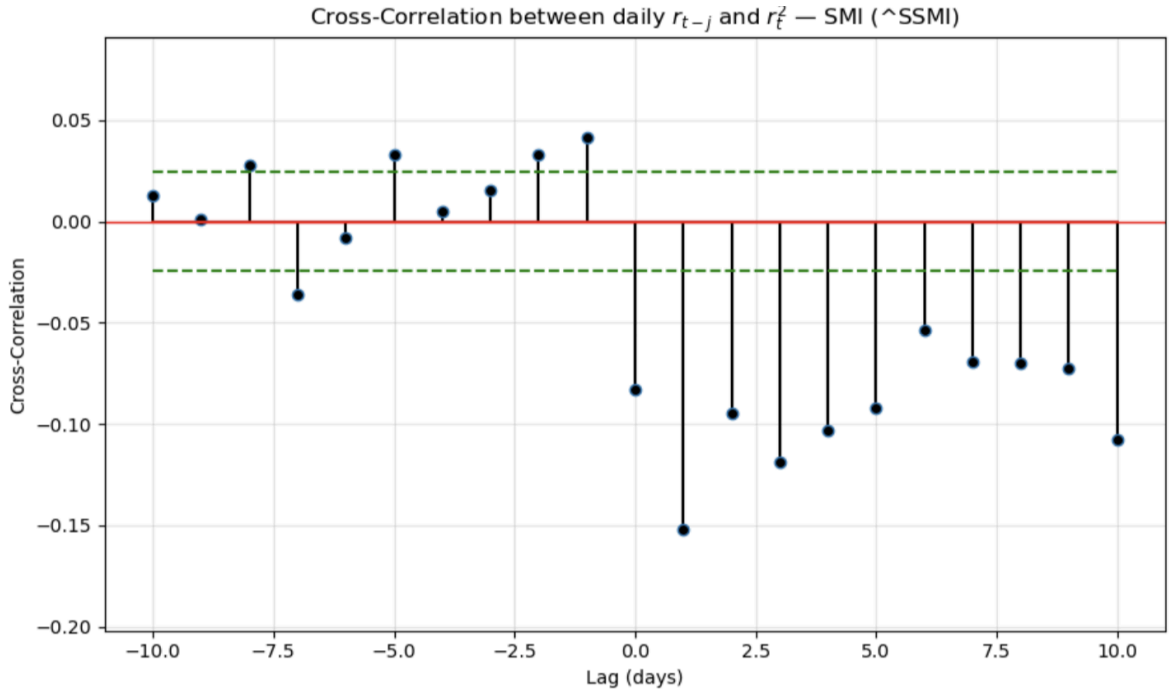
This is also true at the weekly frequency because ACF ≥ 0, but to a lesser extent because the autocorrelation function decreases massively (the lingering effect fades much faster). And at the monthly frequency it's not true anymore because the sign changes.
This indicates that the volatility clustering effect is mostly witnessed at a day scale, and not at a month scale. In other words, the lingering typically lasts a few days, not a few months.



The rolling standard deviation plot shows moments of grouped massive volatility (2001-2004, 2008-2010, 2020-2022) and calm periods grouped together. This is consistent with the volatility clustering.

Fact #8 (Leverage Effect):
We want to demonstrate: When prices fall, volatility tends to rise because investors perceive it as riskier (because market cap thus goes down, therefore a widely-used valuing ratio drops and they're less willing to pay)

This graph shows the correlation between current magnitude of returns (squared returns) and the previous returns. For all lags ≥ 0 (the right-hand side of the graph), all cross-correlations are negative. This illustrates the leverage effect: negative past returns are associated with higher future volatility, with squared returns serving as a proxy for volatility.



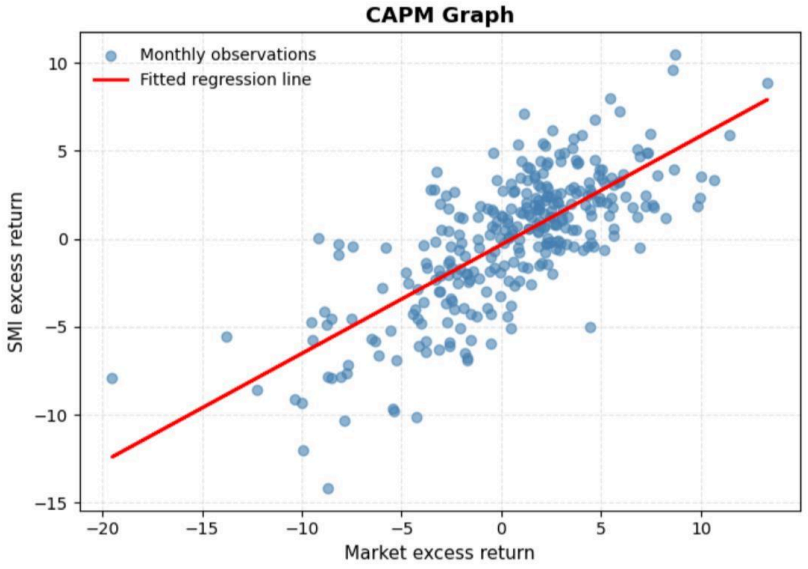
Question 2:

Note: here we used the SMI (Switzerland) as asset and the S&P 500 (U.S.) as market excess return proxy. But to be rigorous, an asset should only be compared to the index of its region, such as Berkshire Hathaway & the S&P 500, or Hermès & the CAC40.

	CAPM (SMI)
const	−0.3292**
	(0.1479)
Mkt-RF	0.6181***
	(0.0326)
R-squared	0.5377
R-squared Adj.	0.5362
N	311

Standard errors in parentheses.
* p<.1, ** p<.05, ***p<.01

The CAPM equation is:
asset excess return = α + β*market excess return.
Here: **SMI excess return = -0.33 + 0.62*market excess return**, where the market is the S&P 500.
MKT-RF has three asterisks, meaning p < 0.01 => **at 95%, this beta is statistically significant**, and is therefore statistically different from 0 (and has an impact). It's even still true at 99%.



	FF3
const	−0.3578**
	(0.1433)
Mkt-RF	0.6398***
	(0.0318)
SMB	−0.3398***
	(0.0728)
HML	0.0658
	(0.0510)
R-squared	0.5735
R-squared Adj.	0.5693
N	311

Standard errors in parentheses.
* p<.1, ** p<.05, ***p<.01

As for the **3-factor model**, it's typically written as:
asset excess return = α + β_market*MKT + β_SMB*SMB + β_HML*HML
Here: **SMI excess return = -0.36 + 0.64*MKT - 0.34*SMB + 0.07*HML**
MKT and SMB βs all have three asterisks, meaning they have p < 0.01 => **at 95%, MKT and SMB are statistically significant**, and are therefore statistically different from 0 (and it's even true at 99%).
However, for HML, t = value/SE = 0.0658/0.0510 ≈ 1.29 < 1.96. So at 95%, **HML is not statistically significant** and can be equal to 0.

Sheer size of the betas:

0.62 is reasonable because it means that the SMI moves on average by 0.62% for a 1% move in the S&P 500 (whether positive or negative). This is credible because **Switzerland is fairly isolated and different from the US, but at the same time the same causes produce the same consequences** (geopolitical uncertainty, also central banks often make similar decisions in similar timelines such as rising base interest rates).

As for the 3-factor model, 0.64 is close to 0.62 so this reinforces our first CAPM model. 0.07 is not reliable because it could be equal to zero (statistically insignificant). And -0.34 is expected because the SMI has much smaller companies than the S&P 500, so if the big companies outperform, the S&P will go up (because mostly comprised of big) whereas the SMI might go down (**mostly comprised of small**, and people will divest the shares of small to go toward big).

Accuracy of the models:

The first model (traditional CAPM) has R**2 = 0.54.

The second model (3-factor) has R**2 = 0.57 (because there are multiple linear variables, we decide to use the adjusted R**2 instead to compensate for the natural increased accuracy due to the increase in number of explanatory variables).

Therefore, since the second model has the largest R**2, it is the most accurate, but not by much (only by about 0.03). And actually, **both models aren't satisfactory because their R**2 are only around 0.5**, whereas for a model to be accurate we expect at least R**2 ≥ 0.9, if not ≥ 0.95.

Statistical significance of the α:

Please note that in our other classes, only the traditional CAPM has been addressed, and other models like APT haven't.

	Alpha	t-stat	p-value
CAPM	-0.329215	-2.225869	0.026744
FF3	-35.784318	-2.497748	0.013022

p-value of the α of the traditional CAPM: 0.03

p-value of the α of the 3-factor model: 0.01

Both have $p \leq 0.05$. Therefore, at 95%, both are statistically different from 0.

We know: CAPM ⇔ α = 0. **At 95%, the CAPM doesn't hold, in both cases** (and it's the case even at 99% because p_FF3 > 0.01).